

Practical No. 1: Introduction to Business Analytics

AIM:(a)Collect data from a real-life business scenario and perform Exploratory Data Analysis (EDA).

(B)Analyze customer data to identify trends and patterns for business decision-making.

CODE:

```
print("TYCS T110 SANJANA SAHU")

import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

# Load the dataset

df = pd.read_csv('/content/supermarket_sales - supermarket_sales.csv')

# Explore Dataset

print(df.info())

print(df.head())

# Check for missing values

print(df.isnull().sum())

# Customer Analysis

# Male vs Female Customers

sns.countplot(x='Gender', data=df)

plt.title("Customer Gender Distribution")

plt.show()
```

Practical No. 1: Introduction to Business Analytics

```
# Member vs Normal Customers

sns.countplot(x='Customer type', data=df)

plt.title("Customer Type")

plt.show()

# Payment Method Used

sns.countplot(x='Payment', data=df)

plt.title("Payment Method")

plt.show()

# Sales Analysis

# Product Line Sales

plt.figure(figsize=(10,5))

sns.countplot(y='Product line', data=df)

plt.title("Products Sold")

plt.show()

# Total Sales by City

city_sales = df.groupby('City')['Total'].sum()

city_sales.plot(kind='bar')

plt.ylabel("Sales")

plt.title("Sales by City")

plt.show()

# Monthly Sales Trend

df['Date'] = pd.to_datetime(df['Date'])

df['Month'] = df['Date'].dt.month_name()
```

Practical No. 1: Introduction to Business Analytics

```
monthly = df.groupby('Month')['Total'].sum()

monthly.plot(kind='bar')

plt.title("Monthly Sales")

plt.xlabel("Month")

plt.ylabel("Total Sales")

plt.show()

# Customer Ratings

sns.histplot(df['Rating'], bins=10)

plt.title("Customer Ratings")

plt.show()

# Correlation Heatmap

plt.figure(figsize=(8,6))

sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.show()
```

OUTPUT:

Practical No. 1: Introduction to Business Analytics

```
TYCS T110 SANJANA SAHU
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
#   Column              Non-Null Count  Dtype
---  --
0   Invoice ID           1000 non-null   object
1   Branch              1000 non-null   object
2   City                1000 non-null   object
3   Customer type       1000 non-null   object
4   Gender              1000 non-null   object
5   Product line        1000 non-null   object
6   Unit price          1000 non-null   float64
7   Quantity            1000 non-null   int64
8   Tax 5%              1000 non-null   float64
9   Total               1000 non-null   float64
10  Date                1000 non-null   object
11  Time                1000 non-null   object
12  Payment             1000 non-null   object
13  cogs                1000 non-null   float64
14  gross margin percentage 1000 non-null   float64
15  gross income         1000 non-null   float64
16  Rating              1000 non-null   float64
dtypes: float64(7), int64(1), object(9)
memory usage: 132.9+ KB
None
Invoice ID Branch      City Customer type Gender \
0  750-67-8428      A  Yangon      Member  Female
1  226-31-3081      C  Naypyitaw  Normal  Female
2  631-41-3108      A  Yangon      Normal  Male
3  123-19-1176      A  Yangon      Member  Male
4  373-73-7910      A  Yangon      Normal  Male

Product line Unit price Quantity Tax 5% Total      Date \
0   Health and beauty  74.69      7  26.1415  548.9715  1/5/2019
1  Electronic accessories  15.28      5   3.8200   80.2200   3/8/2019
2   Home and lifestyle  46.33      7  16.2155  340.5255  3/3/2019
3   Health and beauty  58.22      8  23.2880  489.0480  1/27/2019
4   Sports and travel  86.31      7  30.2085  634.3785  2/8/2019

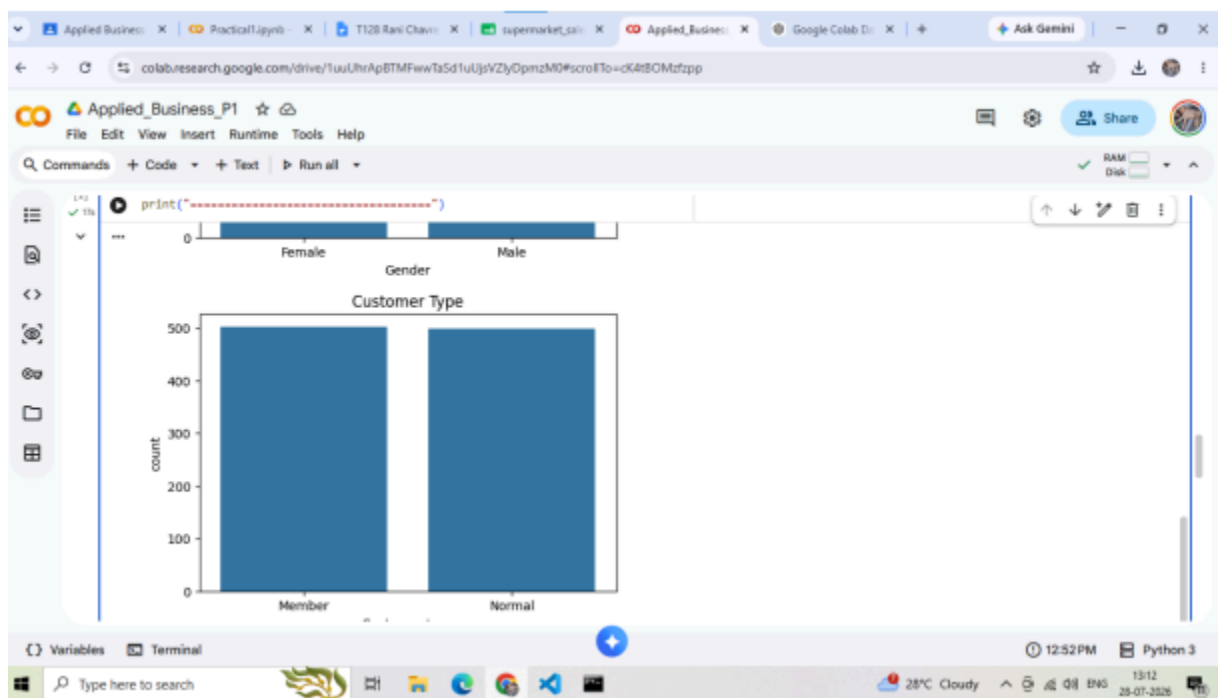
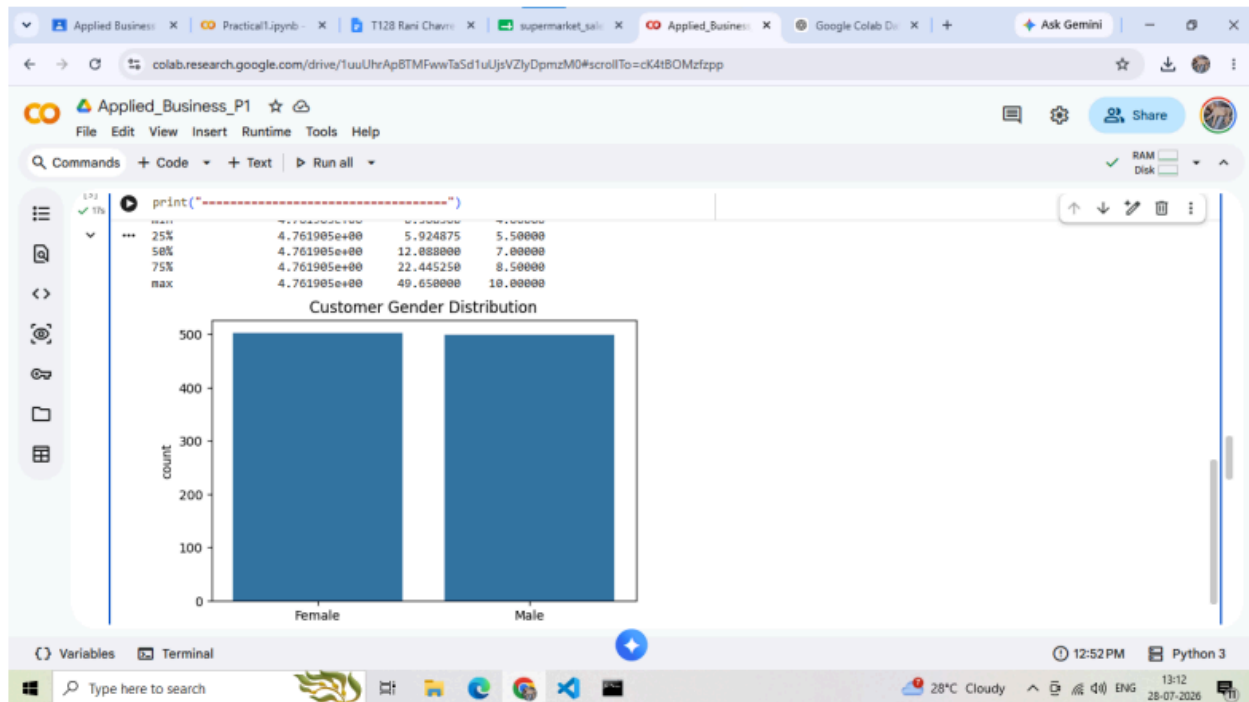
Time Payment cogs gross margin percentage gross income Rating
0 13:08 Ewallet 522.83 4.761905 26.1415 9.1
1 10:29 Cash 76.40 4.761905 3.8200 9.6
2 13:23 Credit card 324.31 4.761905 16.2155 7.4

11 Time 1000 non-null object
12 Payment 1000 non-null object
13 cogs 1000 non-null float64
14 gross margin percentage 1000 non-null float64
15 gross income 1000 non-null float64
16 Rating 1000 non-null float64
dtypes: float64(7), int64(1), object(9)
memory usage: 132.9+ KB
None
Invoice ID Branch      City Customer type Gender \
0  750-67-8428      A  Yangon      Member  Female
1  226-31-3081      C  Naypyitaw  Normal  Female
2  631-41-3108      A  Yangon      Normal  Male
3  123-19-1176      A  Yangon      Member  Male
4  373-73-7910      A  Yangon      Normal  Male

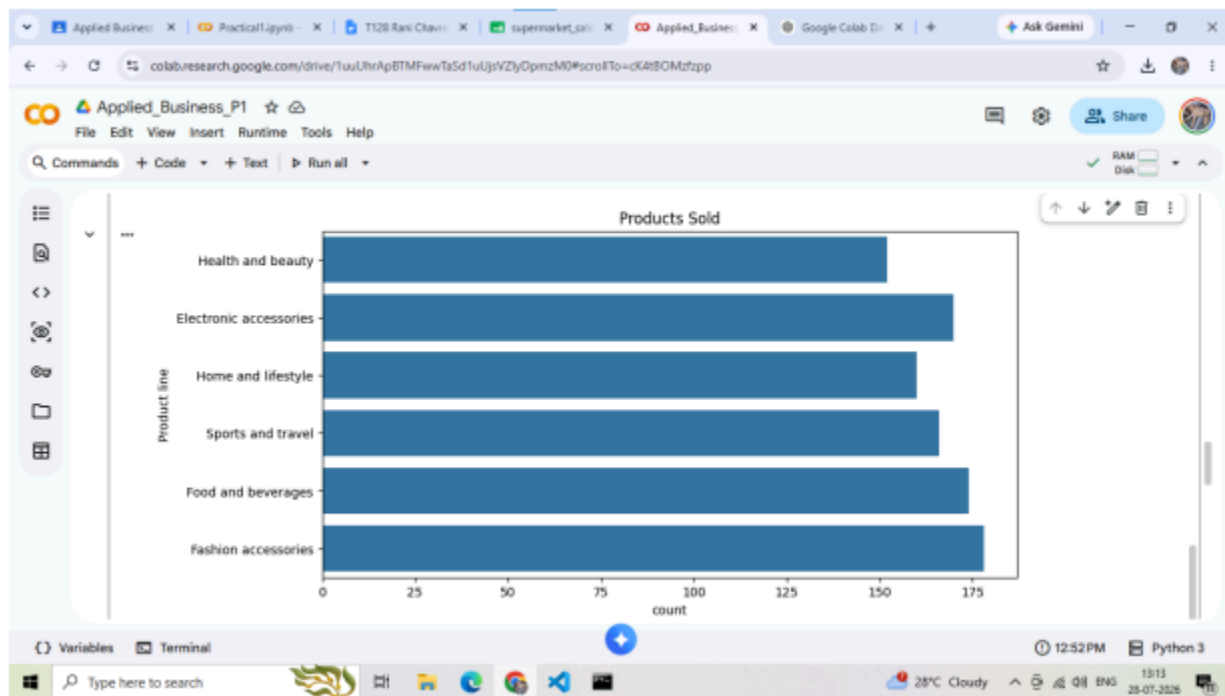
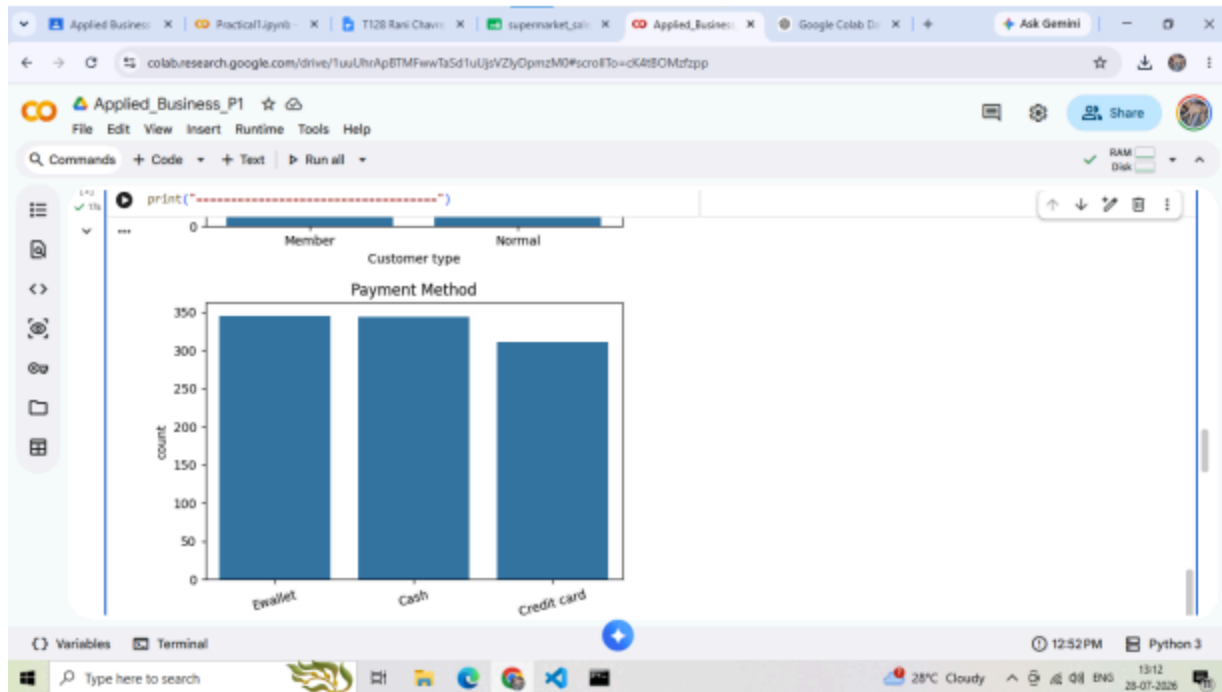
Product line Unit price Quantity Tax 5% Total      Date \
0   Health and beauty  74.69      7  26.1415  548.9715  1/5/2019
1  Electronic accessories  15.28      5   3.8200   80.2200   3/8/2019
2   Home and lifestyle  46.33      7  16.2155  340.5255  3/3/2019
3   Health and beauty  58.22      8  23.2880  489.0480  1/27/2019
4   Sports and travel  86.31      7  30.2085  634.3785  2/8/2019

Time Payment cogs gross margin percentage gross income Rating
0 13:08 Ewallet 522.83 4.761905 26.1415 9.1
1 10:29 Cash 76.40 4.761905 3.8200 9.6
2 13:23 Credit card 324.31 4.761905 16.2155 7.4
3 20:33 Ewallet 465.76 4.761905 23.2880 8.4
4 10:37 Ewallet 604.17 4.761905 30.2085 5.3
Invoice ID 0
Branch 0
City 0
Customer type 0
Gender 0
Product line 0
Unit price 0
Quantity 0
Tax 5% 0
Total 0
Date 0
Time 0
Payment 0
cogs 0
gross margin percentage 0
```

Practical No. 1: Introduction to Business Analytics



Practical No. 1: Introduction to Business Analytics



Practical No. 1: Introduction to Business Analytics

